

ML / DL INTERNSHIP · ACIBADEM MAAÜ · 2026

Predicting Village Wealth from Space

A modern replication – and fairness & uncertainty audit – of Yeh et al. (2020),
Nature Communications

MOTIVATION

Why Predict Poverty from Space?

THE PROBLEM

- ▶ Reliable poverty data is scarce, expensive, and infrequent
- ▶ Household surveys cost millions and run every ~5 years
- ▶ Thinnest exactly where the need is greatest

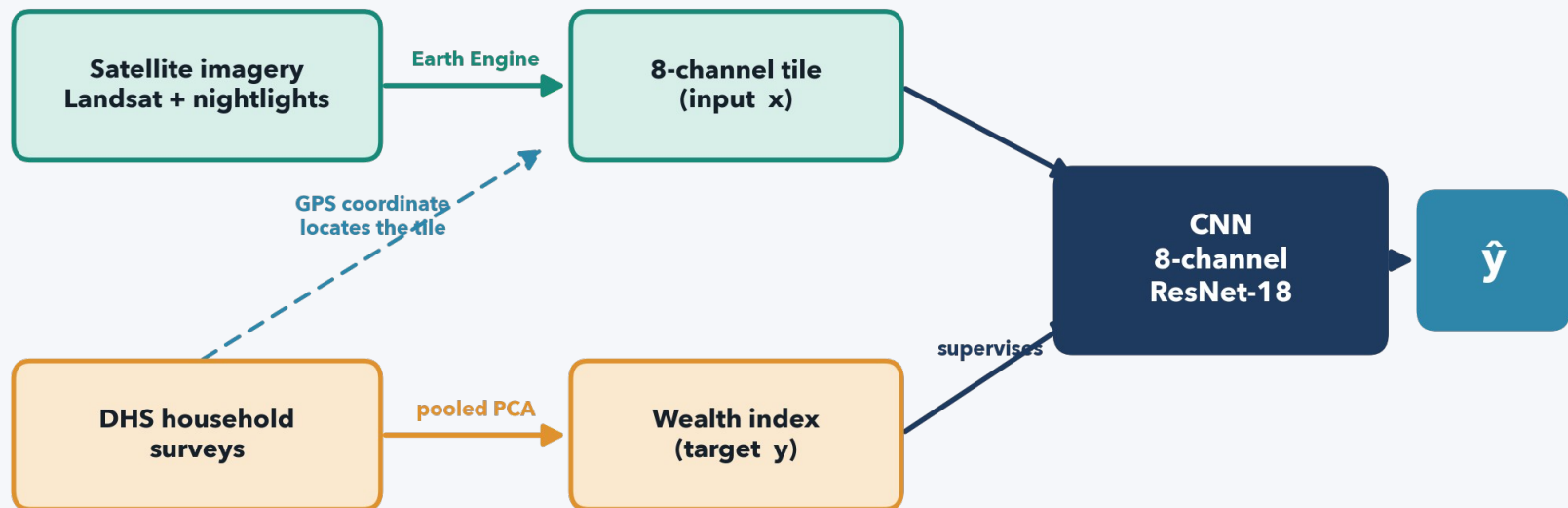
THE PROMISE

- ▶ Satellites image everywhere on Earth – free and repeatedly
- ▶ Can a neural network read village wealth from imagery?
- ▶ If yes: a scalable complement to ground surveys

The whole question: can a free satellite image stand in for an expensive survey?

THE IDEA

One Model, Two Inputs



A village's GPS pulls a satellite tile (x); its survey gives a wealth value (y); the network learns $x \rightarrow y$.

This talk answers two things: where does the wealth number come from, and does it work – for whom?

The Answer Key: DHS Surveys

WHAT DHS GIVES US

- ▶ Gold-standard household surveys (USAID-funded)
- ▶ Unit = a cluster $\approx$ a village (~25 households)
- ▶ A GPS coordinate for every cluster

THE DELIBERATE TWIST

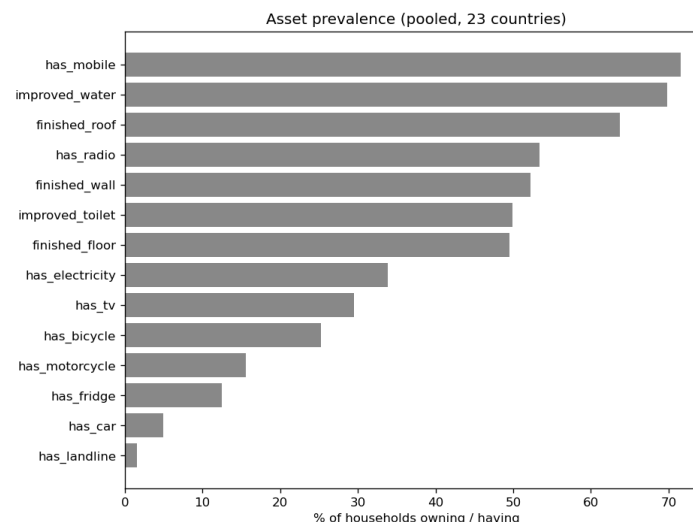
- ▶ DHS shifts each GPS point for privacy: up to 2 km (urban), 5 km (rural)
- ▶ So the true village is somewhere nearby – not exactly on the point
- ▶ That is WHY our tile is a wide 6.72 km box: big enough to still contain it

The survey is our 'answer key' – the model is trained and graded against it.

There Is No 'Wealth' Column

THE PROBLEM

- ▶ DHS never asks 'how rich are you?'
- ▶ It records asset checkboxes: electricity? TV? fridge? car?
- ▶ Plus finished floor / wall / roof, water, toilet, rooms
- ▶ We must turn ~15 yes/no answers into ONE wealth number

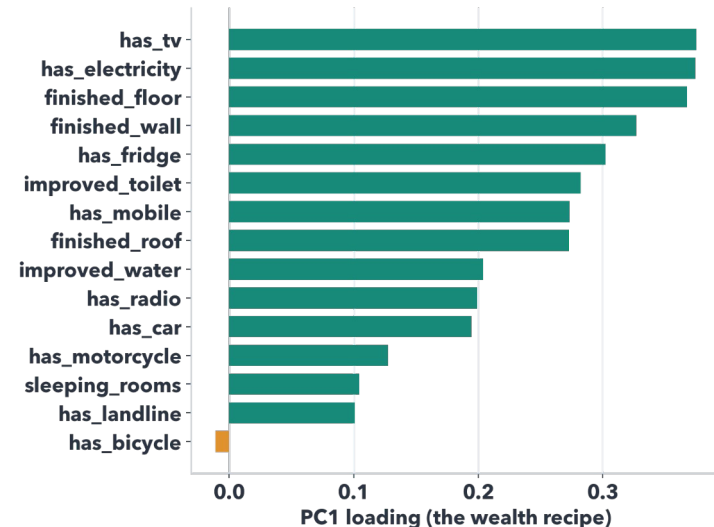


Asset ownership across our 355,000 households – the raw material for the wealth index.

PCA: Finding the Hidden Wealth Axis

THE ONE RULER

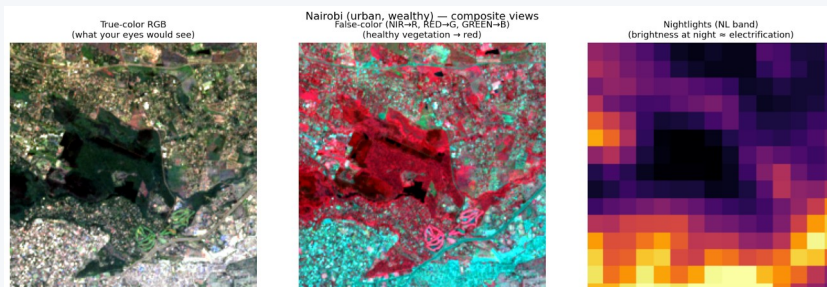
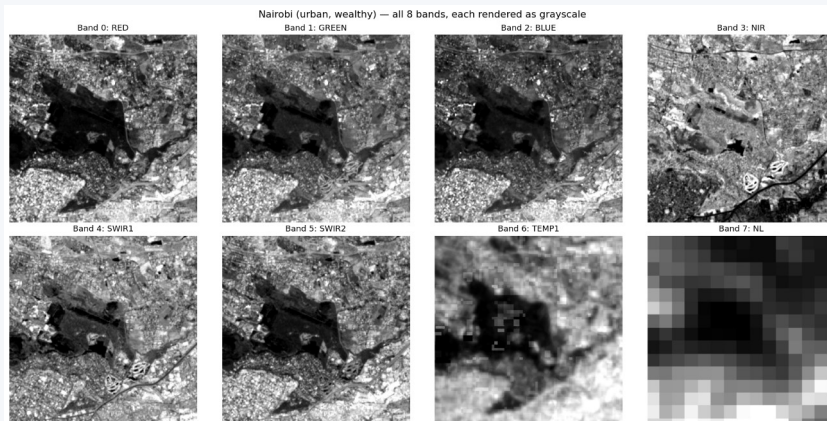
- ▶ PCA finds the single mix of assets that best explains who owns MORE
- ▶ That axis (PC1) is wealth – every asset loads positive (a 'more-of-everything' axis)
- ▶ Average it to the village = our target y
- ▶ 'Only 28%?' – that is $4\times$ the random floor; validity is in the loadings, not the %



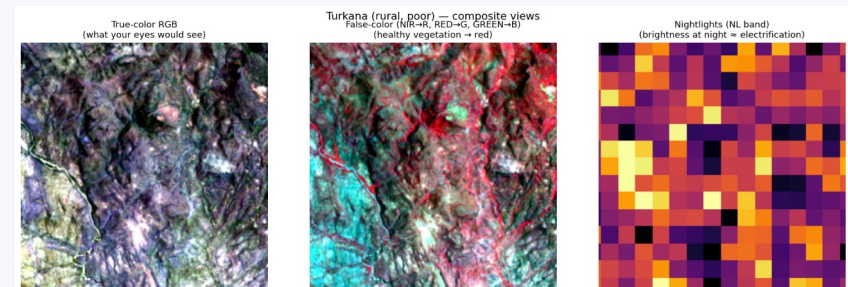
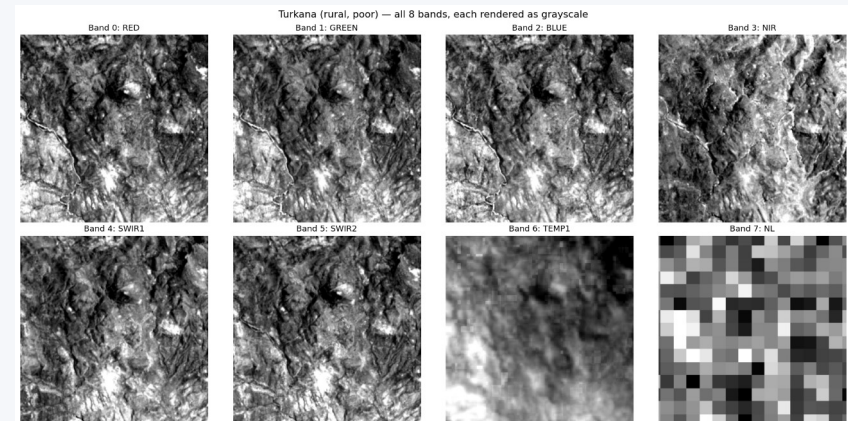
Like a credit-score formula – discovered from the data, not hand-set.

Eight Ways to See a Village

URBAN – NAIROBI, KENYA



RURAL – TURKANA, KENYA



8 channels = 8 'senses': visible colour, infrared (vegetation & rooftops), thermal (heat), night-lights (electrification).
The CNN reads all 8 together and outputs one wealth number.

The Model, in 20 Seconds

- ▶ A ResNet-18 – a standard, well-understood convolutional network
- ▶ Two tweaks: the input is rebuilt for 8 channels (not 3), and the output is one number (not categories)
- ▶ Trained from scratch on the satellite tiles

The science is in the data and the honest evaluation – not the architecture.

Testing on Countries It Has Never Seen

WHY NOT A RANDOM SPLIT?

- ▶ Split villages randomly and the model can MEMORISE a country
- ▶ Same country in train AND test = it cheats – a flattering but fake score
- ▶ Useless for the real question: does it generalise to a NEW place?

CROSS-COUNTRY VALIDATION

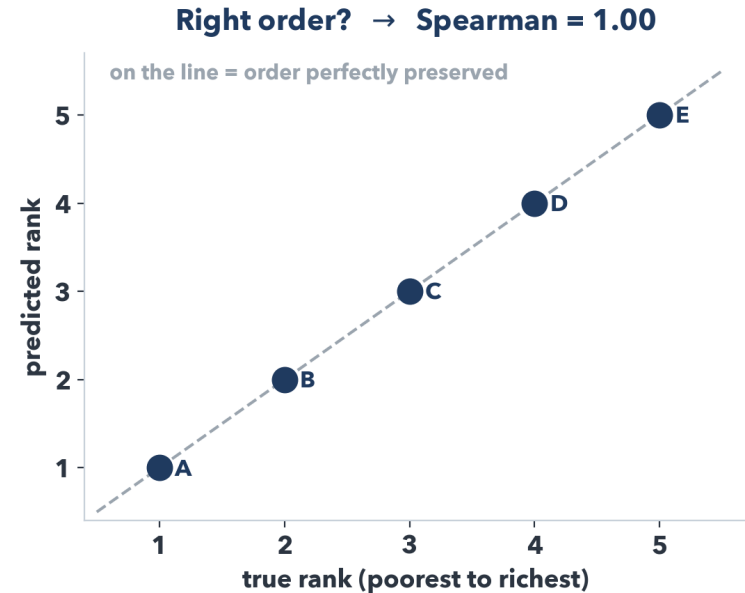
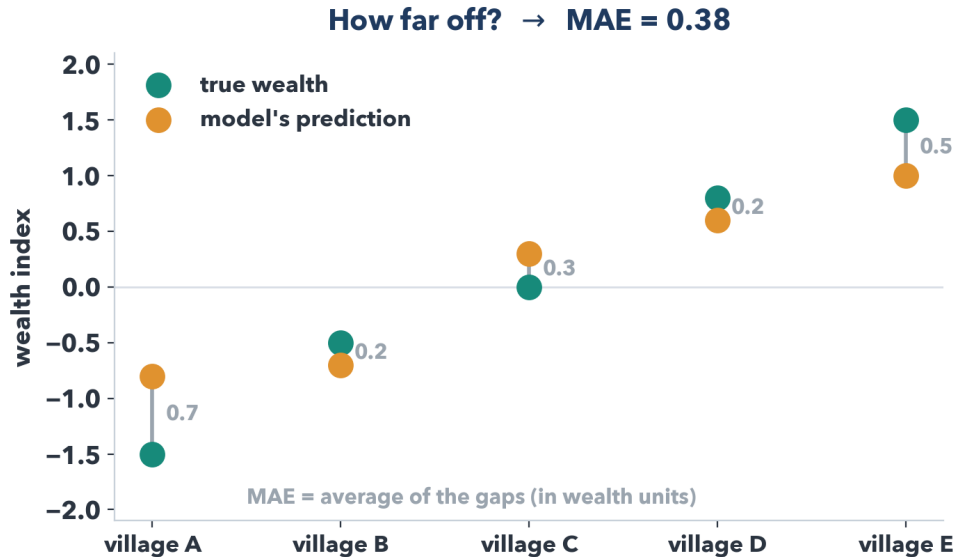
- ▶ Train on ~18 countries, test on 5 it has NEVER seen
- ▶ Rotate so every country is the test set exactly once
- ▶ Every number we report is on unseen countries

Like an exam with questions you never studied – the only fair test.

How Do We Know If It's Good?

Three metrics – because they answer different questions.

Same 5 villages: predictions are imperfect (MAE 0.38) – but perfectly ordered (Spearman 1.00)



$r^2 = 0.83$ · For finding the poorest, ORDER is what matters – a model can rank perfectly even when every value is a little off.

Does It Work? Yes.

0.57

pooled r^2 (ours)

0.52

mean r^2 (ours)

0.67

Yeh 2020 (pooled)

0.25

naïve floor

- ▶ It genuinely reads wealth from space – over 2× a nightlights-only baseline
- ▶ On countries it has never seen (the hard, honest test)
- ▶ The gap to the 0.67 benchmark is DATA, not method: we used one survey round per country; the original pooled many
- ▶ Confirmed: hyperparameter search did not help – we are at the data ceiling, not a tuning ceiling

What Carries the Signal? Night-lights.

0.58

night-lights only

0.52

all 8 channels

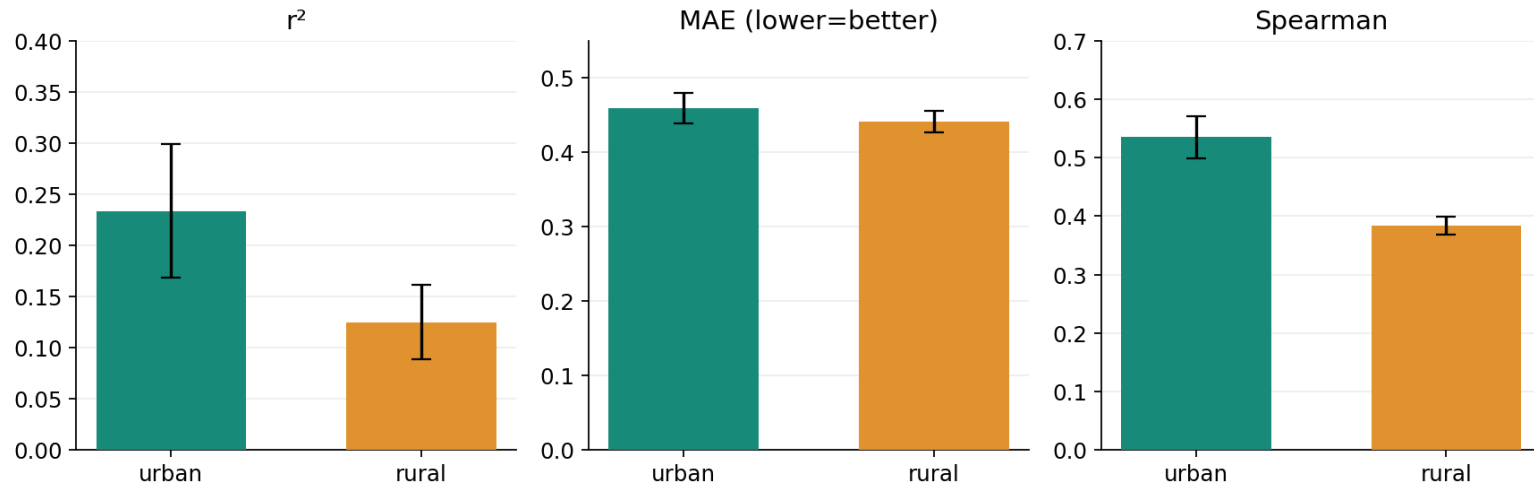
0.46

daytime only

- ▶ Night-lights ALONE roughly match the full 8-channel model
- ▶ Lights = electrification = the single strongest village-wealth signal
- ▶ Daytime imagery adds little here (it is data-hungry, and we are data-limited)
- ▶ Foreshadows the limit: if the signal is mostly lights, what happens where there are none?

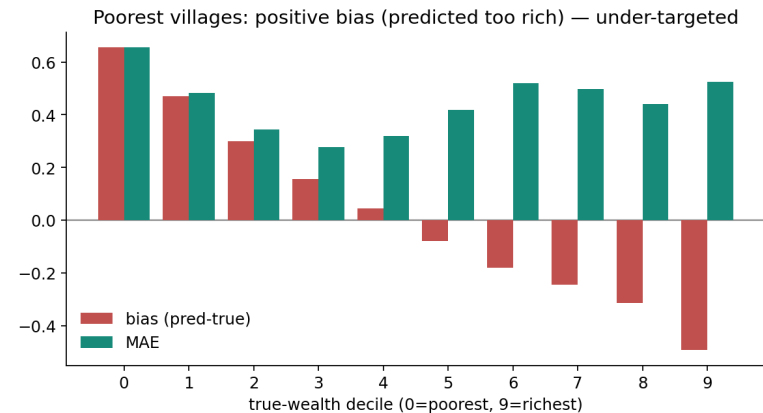
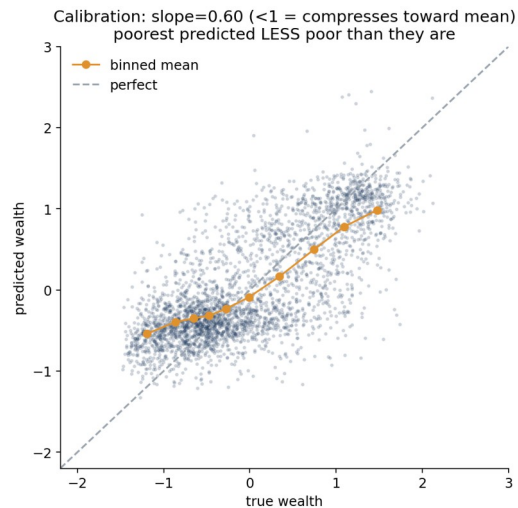
It Serves Cities Better Than Villages

Urban vs Rural — equal absolute accuracy (MAE), worse rural ranking (Spearman)



Remember Nairobi vs Turkana? Equal absolute accuracy (MAE), but the model RANKS rural villages worse (Spearman 0.41 vs 0.54) – in 21 of 23 countries.

Confidently Wrong About the Poorest



The model compresses toward the average: it predicts the POOREST as richer than they are, with the biggest errors exactly there. Targeting the poorest 20%, it finds only ~23% of them.

A Fundamental Limit

CONFIDENTLY WRONG

The audit: largest, most biased errors fall on the poorest villages

UNDETECTABLE

No uncertainty method – ensembles, MC-dropout, learned variance – flags it

UNFIXABLE

Reweighting the loss toward the poor doesn't repair it – it just destabilises training

Because the signal isn't there.

Night-lights are uniformly dark below the poverty line – the satellite cannot separate the extreme poor. No amount of modelling fixes missing information.

From Pilot to Full Study

This was a deliberate single-round PILOT. The full study is already running:

FULL DATASET

- ▶ 3× the data: multiple survey rounds
- ▶ Tests the 'gap is data' claim
- ▶ Extraction in progress now

TEMPORAL DRIFT

- ▶ Train on early years, test on later
- ▶ Does the urban/rural gap widen over time?

OUT-OF-DISTRIBUTION

- ▶ Test on wealthier countries
- ▶ Where does the model break?
- ▶ Frozen model, never retrained

What's New Here

SOLID FOUNDATION

- ▶ Faithful replication – a modern PyTorch port of dead 2015 code
- ▶ The most comprehensive fairness + uncertainty audit on this 23-country data
- ▶ Extends prior fairness work from 10 → 23 countries

MOST NOVEL

- ▶ The equity-framed uncertainty result: uncertainty cannot flag the model's failure on the poorest
- ▶ Apparently unstated in the poverty-mapping literature

Honest positioning: this is an AUDIT, not a new architecture – and we don't claim a state-of-the-art number.

THE ONE THING TO REMEMBER

Satellite poverty maps are great for **ranking regions** – not for last-mile targeting of the extreme poor – and standard uncertainty tools can't warn you when they're wrong about the neediest.

Thank you • Questions?